

Appendix B

Various Euler Angle Transformations

This appendix contains mappings between the 12 sets of Euler angles $\boldsymbol{\theta} = (\theta_1, \theta_2, \theta_3)^T$ and either the direction cosine matrix $[C]$ or the body angular velocity vector $\boldsymbol{\omega}$.

B.1 Direction Cosine Matrix in Terms of the 12 Euler Angle Sets

The direction cosine matrix is defined as

$$\{\hat{\mathbf{b}}\} = [C(\theta_1, \theta_2, \theta_3)]\{\hat{\mathbf{n}}\}$$

Thus, it maps vectors with components in the inertial frame into vectors with components taken in the body frame. The shorthand notations $c\theta_i = \cos \theta_i$ and $s\theta_i = \sin \theta_i$ are used here.

Euler angle set	Direction cosine matrix
1-2-1	$\begin{bmatrix} c\theta_2 & s\theta_2 s\theta_1 & -s\theta_2 c\theta_1 \\ s\theta_3 s\theta_2 & -s\theta_3 c\theta_2 s\theta_1 + c\theta_3 c\theta_1 & s\theta_3 c\theta_2 c\theta_1 + c\theta_3 s\theta_1 \\ c\theta_3 s\theta_2 & -c\theta_3 c\theta_2 s\theta_1 - s\theta_3 c\theta_1 & c\theta_3 c\theta_2 c\theta_1 - s\theta_3 s\theta_1 \end{bmatrix}$
1-2-3	$\begin{bmatrix} c\theta_3 c\theta_2 & c\theta_3 s\theta_2 s\theta_1 + s\theta_3 c\theta_1 & -c\theta_3 s\theta_2 c\theta_1 + s\theta_3 s\theta_1 \\ -s\theta_3 c\theta_2 & -s\theta_3 s\theta_2 s\theta_1 + c\theta_3 c\theta_1 & s\theta_3 s\theta_2 c\theta_1 + c\theta_3 s\theta_1 \\ s\theta_2 & -c\theta_2 s\theta_1 & c\theta_2 c\theta_1 \end{bmatrix}$
1-3-1	$\begin{bmatrix} c\theta_2 & s\theta_2 c\theta_1 & s\theta_2 s\theta_1 \\ -c\theta_3 s\theta_2 & c\theta_3 c\theta_2 c\theta_1 - s\theta_3 s\theta_1 & c\theta_3 c\theta_2 s\theta_1 + s\theta_3 c\theta_1 \\ s\theta_3 s\theta_2 & -s\theta_3 c\theta_2 c\theta_1 - c\theta_3 s\theta_1 & -s\theta_3 c\theta_2 s\theta_1 + c\theta_3 c\theta_1 \end{bmatrix}$
1-3-2	$\begin{bmatrix} c\theta_3 c\theta_2 & c\theta_3 s\theta_2 c\theta_1 + s\theta_3 s\theta_1 & c\theta_3 s\theta_2 s\theta_1 - s\theta_3 c\theta_1 \\ -s\theta_2 & c\theta_2 c\theta_1 & c\theta_2 s\theta_1 \\ s\theta_3 c\theta_2 & s\theta_3 s\theta_2 c\theta_1 - c\theta_3 s\theta_1 & s\theta_3 s\theta_2 s\theta_1 + c\theta_3 c\theta_1 \end{bmatrix}$

Euler angle set	Direction cosine matrix
2-1-2	$\begin{bmatrix} -s\theta_3 c\theta_2 s\theta_1 + c\theta_3 c\theta_1 & s\theta_3 s\theta_2 & -s\theta_3 c\theta_2 c\theta_1 - c\theta_3 s\theta_1 \\ s\theta_2 s\theta_1 & c\theta_2 & s\theta_2 c\theta_1 \\ c\theta_3 c\theta_2 s\theta_1 + s\theta_3 c\theta_1 & -c\theta_3 s\theta_2 & c\theta_3 c\theta_2 c\theta_1 - s\theta_3 s\theta_1 \end{bmatrix}$
2-1-3	$\begin{bmatrix} s\theta_3 s\theta_2 s\theta_1 + c\theta_3 c\theta_1 & s\theta_3 c\theta_2 & s\theta_3 s\theta_2 c\theta_1 - c\theta_3 s\theta_1 \\ c\theta_3 s\theta_2 s\theta_1 - s\theta_3 c\theta_1 & c\theta_3 c\theta_2 & c\theta_3 s\theta_2 c\theta_1 + s\theta_3 s\theta_1 \\ c\theta_2 s\theta_1 & -s\theta_2 & c\theta_2 c\theta_1 \end{bmatrix}$
2-3-1	$\begin{bmatrix} c\theta_2 c\theta_1 & s\theta_2 & -c\theta_2 s\theta_1 \\ -c\theta_3 s\theta_2 c\theta_1 + s\theta_3 s\theta_1 & c\theta_3 c\theta_2 & c\theta_3 s\theta_2 s\theta_1 + s\theta_3 c\theta_1 \\ s\theta_3 s\theta_2 c\theta_1 + c\theta_3 s\theta_1 & -s\theta_3 c\theta_2 & -s\theta_3 s\theta_2 s\theta_1 + c\theta_3 c\theta_1 \end{bmatrix}$
2-3-2	$\begin{bmatrix} c\theta_3 c\theta_2 c\theta_1 - s\theta_3 s\theta_1 & c\theta_3 s\theta_2 & -c\theta_3 c\theta_2 s\theta_1 - s\theta_3 c\theta_1 \\ -s\theta_2 c\theta_1 & c\theta_2 & s\theta_2 s\theta_1 \\ s\theta_3 c\theta_2 c\theta_1 + c\theta_3 s\theta_1 & s\theta_3 s\theta_2 & -s\theta_3 c\theta_2 s\theta_1 + c\theta_3 c\theta_1 \end{bmatrix}$
3-1-2	$\begin{bmatrix} -s\theta_3 s\theta_2 s\theta_1 + c\theta_3 c\theta_1 & s\theta_3 s\theta_2 c\theta_1 + c\theta_3 s\theta_1 & -s\theta_3 c\theta_2 \\ -c\theta_2 s\theta_1 & c\theta_2 c\theta_1 & s\theta_2 \\ c\theta_3 s\theta_2 s\theta_1 + s\theta_3 c\theta_1 & -c\theta_3 s\theta_2 c\theta_1 + s\theta_3 s\theta_1 & c\theta_3 c\theta_2 \end{bmatrix}$
3-1-3	$\begin{bmatrix} c\theta_3 c\theta_1 - s\theta_3 c\theta_2 s\theta_1 & c\theta_3 s\theta_1 + s\theta_3 c\theta_2 c\theta_1 & s\theta_3 s\theta_2 \\ -s\theta_3 c\theta_1 - c\theta_3 c\theta_2 s\theta_1 & -s\theta_3 s\theta_1 + c\theta_3 c\theta_2 c\theta_1 & c\theta_3 s\theta_2 \\ s\theta_2 s\theta_1 & -s\theta_2 c\theta_1 & c\theta_2 \end{bmatrix}$
3-2-1	$\begin{bmatrix} c\theta_2 c\theta_1 & c\theta_2 s\theta_1 & -s\theta_2 \\ s\theta_3 s\theta_2 c\theta_1 - c\theta_3 s\theta_1 & s\theta_3 s\theta_2 s\theta_1 + c\theta_3 c\theta_1 & s\theta_3 c\theta_2 \\ c\theta_3 s\theta_2 c\theta_1 + s\theta_3 s\theta_1 & c\theta_3 s\theta_2 s\theta_1 - s\theta_3 c\theta_1 & c\theta_3 c\theta_2 \end{bmatrix}$
3-2-3	$\begin{bmatrix} c\theta_3 c\theta_2 c\theta_1 - s\theta_3 s\theta_1 & c\theta_3 c\theta_2 s\theta_1 + s\theta_3 c\theta_1 & -c\theta_3 s\theta_2 \\ -s\theta_3 c\theta_2 c\theta_1 - c\theta_3 s\theta_1 & -s\theta_3 c\theta_2 s\theta_1 + c\theta_3 c\theta_1 & s\theta_3 s\theta_2 \\ s\theta_2 c\theta_1 & s\theta_2 s\theta_1 & c\theta_2 \end{bmatrix}$

B.2 Mapping Between Body Angular Velocity Vector and the Euler Angle Rates

The following list provides the various forward and inverse mappings between the body angular velocity vector ω and the Euler angle rates $\dot{\theta} = (\dot{\theta}_1, \dot{\theta}_2, \dot{\theta}_3)^T$. The matrix $[B]$ is defined as

$$\dot{\theta} = [B(\theta)]\omega$$

The shorthand notations $c\theta_i = \cos \theta_i$ and $s\theta_i = \sin \theta_i$ are used again here.

Euler angle set	$[B(\theta)]$	$[B(\theta)]^{-1}$
1-2-1	$\frac{1}{s\theta_2} \begin{bmatrix} 0 & s\theta_3 & c\theta_3 \\ 0 & s\theta_2 c\theta_3 & -s\theta_2 s\theta_3 \\ s\theta_2 & -c\theta_2 s\theta_3 & -c\theta_2 c\theta_3 \end{bmatrix}$	$\begin{bmatrix} c\theta_2 & 0 & 1 \\ s\theta_2 s\theta_3 & c\theta_3 & 0 \\ s\theta_2 c\theta_3 & -s\theta_3 & 0 \end{bmatrix}$
1-2-3	$\frac{1}{c\theta_2} \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 \\ c\theta_2 s\theta_3 & c\theta_2 c\theta_3 & 0 \\ -s\theta_2 c\theta_3 & s\theta_2 s\theta_3 & c\theta_2 \end{bmatrix}$	$\begin{bmatrix} c\theta_2 c\theta_3 & s\theta_3 & 0 \\ -c\theta_2 s\theta_3 & c\theta_3 & 0 \\ s\theta_2 & 0 & 1 \end{bmatrix}$
1-3-1	$\frac{1}{s\theta_2} \begin{bmatrix} 0 & -c\theta_3 & s\theta_3 \\ 0 & s\theta_2 s\theta_3 & s\theta_2 c\theta_3 \\ s\theta_2 & c\theta_2 c\theta_3 & -c\theta_2 s\theta_3 \end{bmatrix}$	$\begin{bmatrix} c\theta_2 & 0 & 1 \\ -s\theta_2 c\theta_3 & s\theta_3 & 0 \\ s\theta_2 s\theta_3 & c\theta_3 & 0 \end{bmatrix}$
1-3-2	$\frac{1}{c\theta_2} \begin{bmatrix} c\theta_3 & 0 & s\theta_3 \\ -c\theta_2 s\theta_3 & 0 & c\theta_2 c\theta_3 \\ s\theta_2 c\theta_3 & c\theta_2 & s\theta_2 s\theta_3 \end{bmatrix}$	$\begin{bmatrix} c\theta_2 c\theta_3 & -s\theta_3 & 0 \\ -s\theta_2 & 0 & 1 \\ c\theta_2 s\theta_3 & c\theta_3 & 0 \end{bmatrix}$
2-1-2	$\frac{1}{s\theta_2} \begin{bmatrix} s\theta_3 & 0 & -c\theta_3 \\ s\theta_2 c\theta_3 & 0 & s\theta_2 s\theta_3 \\ -c\theta_2 s\theta_3 & s\theta_2 & c\theta_2 c\theta_3 \end{bmatrix}$	$\begin{bmatrix} s\theta_2 s\theta_3 & c\theta_3 & 0 \\ c\theta_2 & 0 & 1 \\ -s\theta_2 c\theta_3 & s\theta_3 & 0 \end{bmatrix}$
2-1-3	$\frac{1}{c\theta_2} \begin{bmatrix} s\theta_3 & c\theta_3 & 0 \\ c\theta_2 c\theta_3 & -c\theta_2 s\theta_3 & 0 \\ s\theta_2 s\theta_3 & s\theta_2 c\theta_3 & c\theta_2 \end{bmatrix}$	$\begin{bmatrix} c\theta_2 s\theta_3 & c\theta_3 & 0 \\ c\theta_2 c\theta_3 & -s\theta_3 & 0 \\ -s\theta_2 & 0 & 1 \end{bmatrix}$
2-3-1	$\frac{1}{c\theta_2} \begin{bmatrix} 0 & c\theta_3 & -s\theta_3 \\ 0 & c\theta_2 s\theta_3 & c\theta_2 c\theta_3 \\ c\theta_2 & -s\theta_2 c\theta_3 & s\theta_2 s\theta_3 \end{bmatrix}$	$\begin{bmatrix} s\theta_2 & 0 & 1 \\ c\theta_2 c\theta_3 & s\theta_3 & 0 \\ -c\theta_2 s\theta_3 & c\theta_3 & 0 \end{bmatrix}$

Euler angle set	$[B(\theta)]$	$[B(\theta)]^{-1}$
2-3-2	$\frac{1}{s\theta_2} \begin{bmatrix} c\theta_3 & 0 & s\theta_3 \\ -s\theta_2 s\theta_3 & 0 & s\theta_2 c\theta_3 \\ -c\theta_2 c\theta_3 & s\theta_2 & -c\theta_2 s\theta_3 \end{bmatrix}$	$\begin{bmatrix} s\theta_2 c\theta_3 & -s\theta_3 & 0 \\ c\theta_2 & 0 & 1 \\ s\theta_2 s\theta_3 & c\theta_3 & 0 \end{bmatrix}$
3-1-2	$\frac{1}{c\theta_2} \begin{bmatrix} -s\theta_3 & 0 & c\theta_3 \\ c\theta_2 c\theta_3 & 0 & c\theta_2 s\theta_3 \\ s\theta_2 s\theta_3 & c\theta_2 & -s\theta_2 c\theta_3 \end{bmatrix}$	$\begin{bmatrix} -c\theta_2 s\theta_3 & c\theta_3 & 0 \\ s\theta_2 & 0 & 1 \\ c\theta_2 c\theta_3 & s\theta_3 & 0 \end{bmatrix}$
3-1-3	$\frac{1}{s\theta_2} \begin{bmatrix} s\theta_3 & c\theta_3 & 0 \\ s\theta_2 c\theta_3 & -s\theta_2 s\theta_3 & 0 \\ -c\theta_2 s\theta_3 & -c\theta_2 c\theta_3 & s\theta_2 \end{bmatrix}$	$\begin{bmatrix} s\theta_3 s\theta_2 & c\theta_3 & 0 \\ s\theta_2 c\theta_3 & -s\theta_3 & 0 \\ c\theta_2 & 0 & 1 \end{bmatrix}$
3-2-1	$\frac{1}{c\theta_2} \begin{bmatrix} 0 & s\theta_3 & c\theta_3 \\ 0 & c\theta_2 c\theta_3 & -c\theta_2 s\theta_3 \\ c\theta_2 & s\theta_2 s\theta_3 & s\theta_2 c\theta_3 \end{bmatrix}$	$\begin{bmatrix} -s\theta_2 & 0 & 1 \\ c\theta_2 s\theta_3 & c\theta_3 & 0 \\ c\theta_2 c\theta_3 & -s\theta_3 & 0 \end{bmatrix}$
3-2-3	$\frac{1}{s\theta_2} \begin{bmatrix} -c\theta_3 & s\theta_3 & 0 \\ s\theta_2 s\theta_3 & s\theta_2 c\theta_3 & 0 \\ c\theta_2 c\theta_3 & -c\theta_2 s\theta_3 & s\theta_2 \end{bmatrix}$	$\begin{bmatrix} -s\theta_2 c\theta_3 & s\theta_3 & 0 \\ s\theta_2 s\theta_3 & c\theta_3 & 0 \\ c\theta_2 & 0 & 1 \end{bmatrix}$